

SatQuery AI — v2 Technical Report

Published baseline model · SIH26167 · Indian Space Research Organisation (ISRO)

GitHub Repo	github.com/sameelkazi/satquery
HF Adapter	Sameelkazi/satquery-qwen25vl-vrsbench-lora-v2
Research Paper	SatQuery_IEEE_Research_Paper.pdf
Companion Doc	v4 Progress Report (expanded fusion model)

1. Status

TRAINED · VALIDATED · PUBLISHED

This is SatQuery AI's original, completed model. It is trained, its results are validated on a held-out test partition, and it is written up as a research paper. This document exists so that anyone reviewing the project — a judge, a mentor, another researcher — can see exactly what was built, on what data, and with what measured results, without having to take any of it on faith.

2. What v2 is

v2 fine-tunes **Qwen2.5-VL-3B-Instruct** using parameter-efficient LoRA adaptation (rank $r=16$, $\alpha=32$) with 4-bit NormalFloat (NF4) base-weight quantization. Only the language-side attention and MLP projections are updated; the vision tower is kept frozen throughout training. The result is a single-image visual-question-answering and referring-expression-grounding model purpose-built for satellite imagery, small enough to run on consumer-tier hardware.

3. Training data

Dataset	Role	Link
VRSBench	Primary single-image VQA, grounding, captioning corpus for remote sensing	https://github.com/xiang709/VRSBench

4. A real bug that was found and fixed

Early grounding evaluations came back at a mean IoU of 0.0215 — barely above chance. Rather than accept that as a ceiling, the coordinate convention feeding the model was checked directly: academic grounding benchmarks and this vision-language decoder serialize bounding boxes in different axis orders ($[x1,y1,x2,y2]$ versus $[y1,x1,y2,x2]$), and the mismatch was silently corrupting every predicted box. Correcting the serialization recovered localization fidelity to **0.3671 mean IoU** — a $17.1\times$ improvement — with no change to the model or training data itself. This is disclosed here because it is a genuine, verifiable finding, not a tuning trick.

5. Benchmark results

Measured on a strictly held-out, disjoint VRSBench test partition ($N=350$). GPT-4V and GeoChat-7B rows are included as independent points of reference, evaluated the same way.

Model	VRAM	VQA Acc	Grounding mIoU	Grounding Acc@0.5	Latency
GPT-4V (zero-shot)	Cloud API	65.6%	0.2840	32.4%	3.40s
GeoChat-7B (base)	28 GB	60.6%	0.2410	28.1%	4.80s
Qwen2.5-VL-3B (zero-shot)	7.8 GB	68.2%	0.0215*	0.0%*	1.85s
SatQuery AI v2 (ours)	2.4 GB	77.0%	0.3671	44.0%	1.11s

**Unadapted zero-shot baseline before the coordinate-serialization fix described in Section 4.*

6. What this proves, and what it doesn't

- It proves the QLoRA training pipeline, the Kaggle training infrastructure, and the base-model choice all work end-to-end on real, held-out data — not just in theory.
- It is scoped to single-image VQA and grounding only. It does not cover bi-temporal change detection or optical-SAR fusion — that is the explicit subject of v4 (see companion document).
- Results are reported on VRSBench's own held-out partition and are not claimed to generalize to out-of-distribution imagery beyond what VRSBench represents.

7. Reproducing v2

The base foundation model is loaded from its public Hugging Face release at runtime. Only the LoRA adapter weights produced by this project's own fine-tuning are hosted separately (binary weights are intentionally excluded from the git repository):

```
pip install peft → PeftModel.from_pretrained(base_model,
"Sameelkazi/satquery-qwen25vl-vrsbench-lora-v2")
```